DIGITAL SIGNAL CHAIN AND DISCRETE-TIME PI CONTROL OF A COMPACT THERMAL CHAMBER

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Project repository: `https://github.com/GMarquesx/PIDlampada`

ABSTRACT

Digital systems are at the heart of computer engineering: physical quantities are sampled, quantized, processed by algorithms and converted back into actions on the physical world. This paper analyzes this digital chain in a compact thermal chamber controlled by an ESP32. The temperature measured by an LM35 sensor is sampled by a 12-bit analog-to-digital converter, filtered by a moving average, processed by a discrete-time proportional-integral (PI) controller and converted into heating power by a 10-bit pulse-width modulation (PWM) signal. The operating modes and safety conditions are organized as a finite state machine. The resolution of each stage, the effect of the digital filter and the discrete implementation of the controller are discussed. Analytically, the ADC step corresponds to a temperature resolution between 0.027 and 0.095 °C, depending on the attenuation setting, the 16-sample moving average is expected to reduce uncorrelated noise by a factor of four, and the PWM resolves about 15 mW of heating power.

Key words *– Digital systems, Quantization, Digital filter, Discrete-time control, Finite state machine.*

1. INTRODUCTION

Most modern control and instrumentation systems are digital: sensors produce continuous signals, but the decisions are taken by processors that work with discrete samples and finite-length binary numbers [1]. Converting a continuous physical process into this digital domain introduces sampling, quantization and finite resolution in both the measurement and the actuation, and the control law itself becomes a difference equation executed at fixed time intervals [2, 3].

This paper uses a compact thermal chamber as a case study of a complete digital system. The chamber is an acrylic box heated by three 12 V/5 W automotive lamps and controlled by an ESP32 microcontroller [4]. Instead of focusing on the hardware, the analysis follows the signal through each digital stage, from the analog-to-digital conversion to the PWM output, and describes the finite state machine that organizes the operation of the system.

2. MATERIAL AND METHODS

2.1. Digital signal chain

Figure 1 shows the path of the signal. The LM35 sensor converts temperature into voltage (10 mV/°C) [5]; the ADC samples and quantizes this voltage; a moving-average filter reduces noise; the discrete PI computes the control action; and the PWM converts the resulting digital duty cycle into average power on the lamps through a MOSFET switch [6].

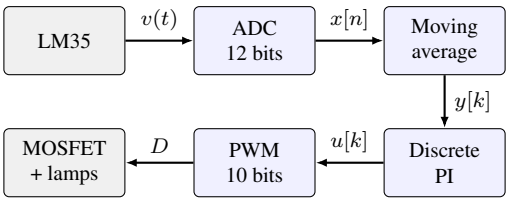


Figure 1: Digital signal chain (gray: analog domain; blue: digital domain).

2.2. Sampling and quantization

The ESP32 ADC has 12 bits, that is, $2^{12} = 4096$ levels. For a full-scale voltage V_{FS}, the quantization step is

$$q = \frac{V_{FS}}{2^{12}} \tag{1}$$

and, since the LM35 produces 10 mV/°C, the temperature resolution is $q\,/\,(10\,\text{mV/°C})$. The ESP32 ADC offers four attenuation settings, with nominal full-scale voltages of 1.1, 1.5, 2.2 and 3.9 V [7]; the resulting steps are listed in Table 1. Since an LM35 output of 1.1 V corresponds to 110 °C, the smallest attenuation that covers the expected temperature range gives the finest resolution. Because the thermal time constant of the chamber is long, a sampling period of 60 s (1 s in bench mode) is enough to represent the temperature without aliasing of the useful signal.

2.3. Moving-average filter

Each temperature sample $y[k]$ is the average of $N = 16$ consecutive conversions:

$$y[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \tag{2}$$

For uncorrelated noise, averaging N samples reduces the noise standard deviation by a factor of $\sqrt{N}$, i.e. by 4 for $N = 16$ [3].

2.4. PWM as a digital-to-power converter

The PWM is generated at 500 Hz with 10-bit resolution, so the duty cycle D has 1024 steps of approximately 0.1 %. With three 5 W lamps (15 W nominal), each step corresponds to roughly 15 mW of average heating power. The MOSFET operates only as an on/off switch, which makes the actuation stage itself a digital circuit.

2.5. Discrete-time PI controller

The continuous PI law is discretized with a sampling period T_s using the backward-Euler approximation of the integral [2]:

$$u[k] = K_p\, e[k] + K_i T_s \sum_{i=0}^{k} e[i] \tag{3}$$

The output is limited to the PWM range (0 to 1023), and the accumulated sum is frozen while the output is saturated to prevent integral windup [8]. Finally, the duty cycle is corrected by the battery voltage, since the lamp power depends on the square of the supply:

$$D_{corr} = D_{PI} \left(\frac{12}{V_{bat}} \right)^2 \tag{4}$$

2.6. Finite state machine

The behavior of the system is organized as the finite state machine of Figure 2. After initialization, the system runs in bench mode (Wi-Fi on, 1 s period) or embedded mode (Wi-Fi off, light sleep, 60 s period). In any state, a temperature above the firmware safety limit forces the FAULT state, which turns the PWM off.

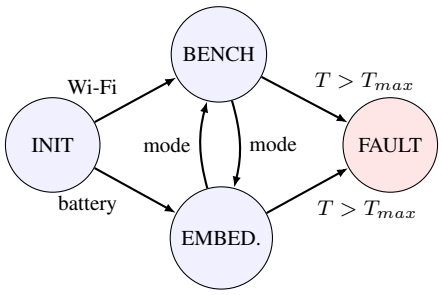


Figure 2: Finite state machine of the chamber firmware.

2.7. Analysis procedure

The performance of each stage is evaluated analytically. The quantization step is computed from Eq. (1) for each attenuation setting of the ADC; the effect of the filter is estimated from the $\sqrt{N}$ relation for uncorrelated noise; and the power resolution of the actuator follows from the number of PWM steps and the nominal power of the lamps.

3. RESULTS

Table 1 gives the quantization step of the ADC for each attenuation setting, computed with Eq. (1) and the nominal full-scale voltages of the ESP32 [7].

The step grows with the attenuation because the same 4096 levels are spread over a larger range: from 0.027 °C at 0 dB to 0.095 °C at 11 dB, a factor of about 3.5.

Regarding the digital filter, averaging $N = 16$ conversions is expected to reduce the standard deviation of uncorrelated noise by a factor of $\sqrt{16} = 4$, at the cost of sixteen conversions per temperature reading. The PWM stage has 1024 duty-cycle steps of approximately 0.1 %, which correspond to about 15 mW of heating power for the three 5 W lamps.

Table 1: ADC step and temperature resolution for each attenuation setting.

Attenuation (dB)	V_{FS} (V)	q (mV)	**Resolution** (°C)
0	1.1	0.269	0.027
2.5	1.5	0.366	0.037
6	2.2	0.537	0.054
11	3.9	0.952	0.095

4. DISCUSSION

The ADC step is the smallest change the converter can represent, while the noise sets how much consecutive readings fluctuate. When the noise is larger than the step, it is the limiting factor, and averaging is effective because the noise itself dithers the quantizer; when it is much smaller than the step, averaging cannot recover resolution. The ESP32 ADC is reported to be sensitive to noise [7], which justifies the 16-sample average. Averaging, however, does not remove systematic errors: the nonlinearity of the ADC [4], the variation of its reference voltage between chips, which is nominally 1.1 V but may range from 1.0 to 1.2 V [7], and the tolerance of the LM35 [5] are not affected by the filter. Beyond a certain point, therefore, the accuracy of the measurement, rather than its resolution, limits the system.

The 60 s sampling period is adequate as long as the dominant time constant of the chamber is well above it, a common guideline being to take several samples within the rise time of the response [2]. The cost of a long period is a slower reaction to disturbances, since the temperature is read only once per period; the benefit is that, in embedded mode, the microcontroller stays in light sleep for most of the time.

A finer PWM would reduce the quantization of the control action, which matters most near the setpoint, where the PI output changes by small amounts. However, the 15 mW step is already small compared with the 15 W nominal power and is smoothed by the thermal inertia of the chamber, so the benefit would be marginal. Likewise, the backward-Euler integration of Eq. (3) is simple and robust, and the trapezoidal (Tustin) rule gives a more accurate integral mainly when the dynamics are fast relative to T_s [2]; the difference between the two methods vanishes as T_s becomes small compared with the time constant of the chamber.

The finite state machine makes the behavior of the firmware explicit: each mode is a state, each change of mode is a transition with a single triggering condition, and the over-temperature transition applies to every operating state, so a fault cannot be masked by a mode change. In the FAULT state the PWM is off regardless of the PI output, which makes this safety behavior simple to verify by test.

Since the firmware only checks the temperature once per sampling period, it is complemented by protections that do not depend on the code, namely a fuse and a thermal cut-off in the power path.

5. CONCLUSIONS

This paper analyzed a compact thermal chamber as a complete digital system, from the quantization of the sensor signal to the PWM actuation, including digital filtering, a discrete-time PI controller and a finite state machine.

The analysis showed that the 12-bit ADC resolves between 0.027 and 0.095 °C per step depending on the attenuation setting, that the $N = 16$ moving average is expected to reduce uncorrelated noise by a factor of four, and that the 10-bit PWM controls the heating power in steps of about 15 mW. The finite state machine organizes the operating modes and the over-temperature protection of the firmware.

Future work includes measuring the noise before and after the filter and the closed-loop response of the discrete PI controller to validate these estimates, calibrating the ESP32 ADC against a reference thermometer or replacing it by an external converter of higher resolution, and comparing the backward-Euler and Tustin discretizations experimentally.

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